

Exercise 4 - SQL Fundamentals (Joins, UNION, Filtering, and Aggregates)

(01) SQL Fundamentals = SELECT and filtering

(01) SELECT *
FROM employees

Id	first-name	department	last-name	Salary	hire-date	city
1	Alice	IT	Steen	70 000	2020-01-01	Johannesburg
2	Brian	HR	Lee	45 000	2019-03-22	Cape Town
3	Cathy	Finance	Zulu	65 000	2018-07-15	Durban
4	David	Marketing	Mokoena	50 000	2021-11-05	Pretoria
5	Eva	IT	Maidoo	22 000	2017-09-08	Johannesburg

(02) SELECT DISTINCT
Department
FROM Employees;

Department
IT
HR
Finance
Marketing

(03) SELECT
first-name
last-name
FROM Employees
ORDER BY Salary IN DESC;

first-name	last-name
Eva	Maidoo
David	Mokoena
Cathy	Zulu
Brian	Lee
Alice	Green

(04) SELECT

id,
first-name,
last-name,
salary

FROM Employees

~~WHERE salary >~~ WHERE salary = MAX(salary) As the highest paid employee

~~ORDER BY DESC~~

~~LIMIT 3;~~

id	first-name	last-name	salary
5	Eva	Maidoo	72000
4	David	Mokoena	5
1	Alice	Green	70000
3	Cathy	Zulu	65000

(02)

(05) SELECT

id,
first-name,
last-name,
department

FROM employees

WHERE department = 'IT'

Id	first-name	Last-name	department
1	Alice	Green	IT
2	Eva	Naidoo	IT

(06) SELECT

Id,
first-name,
last-name,
department,
Salary

FROM Employees

WHERE department = 'Finance' AND Salary > 60000;

Id.	first-name	last-name	department	Salary
2	Cathy	Zulu	Finance	65000

(07) SELECT

Id,
first-name,
last-name,
Department

FROM Employee,

WHERE Department = 'HR' OR 'Marketing'

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

(8) SELECT

id,
first-name,
last-name,
department

FROM Employees

WHERE department $\neq$ IT

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

(9) SELECT

id,
first-name,
last-name,
department

FROM Employees

WHERE department IN (IT, HR, Finance);

Id	First-name	Last-name	department
1	Alice	Steen	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Eva	Naidoo	IT

(10) SELECT

Id,
first-name,
Last-name,
department ~~Salary~~,
Salary,
City

FROM Employees

WHERE department = 'IT' AND Salary > 65000 AND City = 'Johannesburg'

Id	first-name	Last-name	department	Salary	City
1	Alice	Steen	IT	70000	Johannesburg
2	Eva	Naidoo	IT	72000	Johannesburg